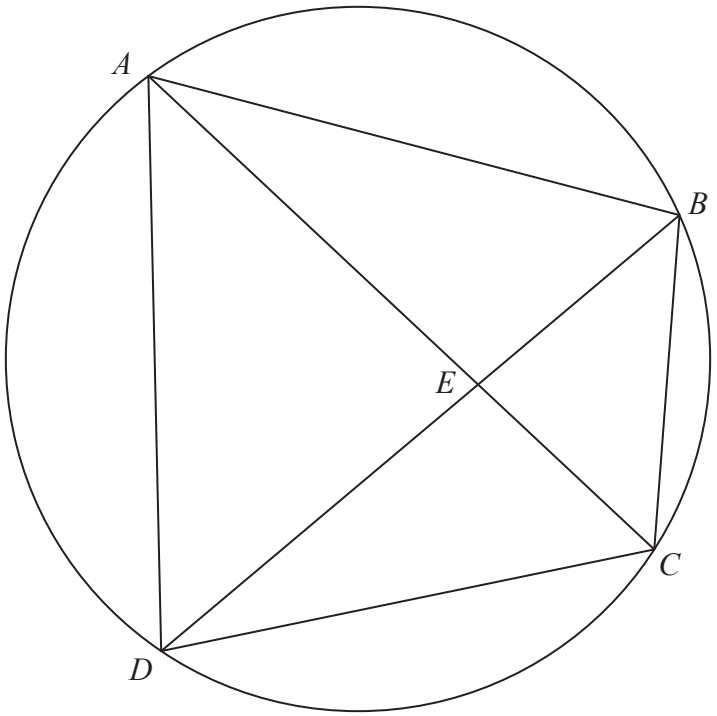
15 A, B, C and D are four points on the circumference of a circle.



AEC and BED are straight lines.

Prove that triangle ABE and triangle DCE are similar.
You must give reasons for each stage of your working.

angle BAE = angle CDE

Angles in the same segment, standing on chord BC.

1 mark
First pair with
a correct reason

angle AEB = angle DEC

Vertically opposite angles.

1 mark
Second pair with
a correct reason

angle ABE = angle DCE

Angles in the same segment, standing on chord AD.

1 mark
All three pairs
with correct reasons

Therefore triangle ABE is similar to triangle DCE (AAA).

(Total for Question 15 is 3 marks)